

CURRICULUM VITAE (CV)

ALEX R DYTSO

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536 S. Lyman St.
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EDUCATION

- | | | |
|------------|---|----------|
| PhD | University of Illinois at Chicago
<i>Research Interests:</i>
Network Information Theory,
Communicating Theory
<i>Advisors:</i> Natasha Devroye, Daniela Tuninetti | Present |
| MS | University of Illinois at Chicago
<i>Research Interests:</i>
Network Information Theory,
Communicating Theory
<i>Advisors:</i> Natasha Devroye, Daniela Tuninetti | Dec 2014 |
| BS | University of Illinois at Chicago
<i>Major:</i> Electrical Engineering | May 2011 |

HONORS AND AWARDS

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|---|---------|
| Wexler Award of Excellence
<i>Description:</i> Awarded to promising graduate students. | 2011-12 |
| International Engineering Consortium's William L. Everitt Student Award of Excellence
<i>Description:</i> Honors outstanding seniors who have demonstrated an interest in the communications field. | 2011 |
| UIC Engineering Alumni Association Scholarship
<i>Description:</i> Awarded to each year to two students who demonstrate high academic achievements and strong desire to enhance their education by demonstrating leadership skills. | 2010 |
| Irene Jones Scholarship
<i>Description:</i> Support the next generation of engineers - the men and women who will change the world. | 2010 |

RESEARCH EXPERIENCE

University of Illinois at Chicago

2011 to Present

Position: Research Assistant

Advisor Name: Natasha Devroye, Daniela Tuninetti

- Network Information Theory
- Communication Theory
- Six Publications (see below)

University of Illinois at Chicago

Spring 2011

Position

Position: Undergraduate Research Assistant

Advisor Name: Natasha Devroye, Daniela Tuninetti

Research Topics:

- Interference Channel with Partial Codebook Knowledge
- Interference Channel with Cognitive Relay

PUBLICATIONS

Conference Papers

A. Dytso, N. Devroye, and D. Tuninetti, ``**The sum-capacity of the symmetric linear deterministic Complete K-user Z-interference channel,**'' *50th Annual Allerton Conference on Communication, Control, and Computing*, Monticello, IL, October 2012.

A. Dytso, N. Devroye, and D. Tuninetti, ``**On The Capacity of the Symmetric Interference Channel with a Cognitive Relay at High SNR,**'' *International Conference on Communications (ICC)*, Ottawa, June 2012.

A. Dytso, N. Devroye, and D. Tuninetti, ``**On Capacity of Interference Channels with Partial Codebook Knowledge,**'' *International Symposium on Information Theory*, Istanbul, July 2013.

A. Dytso, D. Tuninetti, N. Devroye, ``**On Discrete Alphabets for the Two-user Gaussian Interference Channel with One Receiver Lacking Knowledge of the Interfering Codebook,**'' *Information Theory and Applications Workshop*, invited paper.

A. Dytso, N. Devroye, D. Tuninetti, ``**On Gaussian Interference Channels with Mixed Gaussian and Discrete Inputs,**'' *International Symposium on Information Theory*, Honolulu, July 2011.

Dytso, A., Devroye, N., Tuninetti, D., ``**On Gaussian interference channels with mixed Gaussian and discrete inputs,**'' *Information Theory (ISIT), 2014 IEEE International Symposium on*, vol., no., pp.261,265, June 29 2014-July 4 2014

Dytso, A., Tuninetti, D., Devroye, N. " **On the Two-user Interference Channel with Partial Codebook Knowledge at one Receiver: Symmetric Capacity to within a Gap with PAM inputs,**" *IEEE Information Theory Workshop (ITW), Jerusalem, Apr. 2015*

Journal Papers

A. Dytso, N. Devroye, D. Tuninetti, "**On the Two-user Interference Channel with Partial Codebook Knowledge at one Receiver,**" *Wireless Communications, IEEE Transactions on* , vol.13, no.12, pp.6824,6838, Dec. 2014

A. Dytso, N. Devroye, D. Tuninetti, "**On the Capacity Region of the Two-user Interference Channel with a Cognitive Relay,**" *Information Theory, IEEE Transactions on* , vol.61, no.3, pp.1257,1276

RELATED COURSE WORK

- | | |
|-----------------------------------|---------------------------|
| • Wireless Communications | • Random Processes Theory |
| • Digital Communications | • Information Theory |
| • Detection and Estimation Theory | • Measure Theory |
| • Digital Signal Processing | • Computer Algorithms |
| • Adaptive Digital Filter Theory | |

PRESENTATIONS AND INVITED LECTURES

Poster Presentation, "The sum-capacity of the symmetric linear deterministic Complete K-user Z-interference channel," *International Symposium on Information Theory*, Cambridge, MA, July 2012.

Poster Presentation, "The sum-capacity of the symmetric linear deterministic Complete K-user Z-interference channel," *2012 North American School of Information Theory* Cornell University, Ithaca, NY, June 2012.

Paper Presentation, "The sum-capacity of the symmetric linear deterministic Complete K-user Z-interference channel," Proc. 50th Annu., Allerton Conf. Communication, Control, and Computing, Monticello, IL, Oct. 1-5, 2012

Paper Presentation, "On Capacity of Interference Channels with Partial Codebook Knowledge," *International Symposium on Information Theory*, Istanbul, July 2013.

Paper Presentation, "On the Two-user Interference Channel with Partial Codebook Knowledge at one Receiver," *International Symposium on Information Theory*, Honolulu, July 2013.

Invited Talk, "On the Optimality of Treating Interference as Noise in Competitive Scenarios," University of Maryland College Park, Host: Sennur Ulukus.

Poster Presentation, “Discrete inputs in Gaussian interference networks: performance analysis and approximate optimality,” *Information Theory and Applications Workshop*, San Diego, 2015.

PROFESSIONAL TRAINING

Seminar or Workshop

2012 North American School of Information Theory

Cornell University, Ithaca, NY, June 2012.

Description: Seminars on Network Information Theory

PROFESSIONAL AFFILIATIONS

The National Aeronautics and Space Administration (NASA)

Summer 2014

Goddard Space Flight Center, Maryland

Description: Internship. Error Correction Codes for TDRS satellites.

China Solar & Clean Energy Solutions Inc.

Summer 2011

Beijing, China

Description: Assistant to the main engineer. Performed efficiency of deployment calculations on solar water heaters and solar panels. Developed International Confidence, International Sensitivity and Awareness.

Mathematics, Statistics, and Computer Science Tutoring Center

2007 - 2011

University of Illinois at Chicago

Description: Math Tutor. Assisted students with understanding and solving low and high-level math problems. Formed study groups.

DRO Electric Co.

Summer 2009

Chicago, IL

Description: Repairing Technician. Setting up, installing and repairing Air Conditioning Units.

SH Electric Inc.

Summer 2008

Chicago, IL

Description: Installation Technician. Installation security cameras, speaker systems, fire alarms, light fixtures, wiring of houses. etc.

PROJECTS

Computer Cluster (Beowulf clusters)

Fall 2012

Description: Using high computing capabilities of computer

clusters to simulate and approximate Network Information Problems.

MatLab Projects

2012-Present

Description: Full list of MatLab projects can be viewed on MatLabCentral.com.

Projects include: Channel Equalizers, scalar Kalman Filter, 3-D Extended Kalman Filter, Combinatorial functions, functions that manipulate sequences.

Senior Design Project

2010-2011

Name: Watch-White board

Description: Using Chronos eZ-430 watch to translate white board note into digital format.

RF-Design

Fall 2010

Build Fully Operational superheterodyne receiver.

Micro-Controller Based Design

Fall 2009

(all designs were performed on Motorola 68HC11 micro-controller)

- Stop-Watch/Clock, Count-Down/Up timers
- Four-Function Calculator
- Electronic Flashcards
- Voltmeter

COMMUNITY SERVICE AND VOLUNTEERING

Future City Competition

2014, 2015

Description: Judge from local IEEE section.

Local Community Tutoring Center

2011-2012

2250 w Cortez St, Chicago, IL, 60622

Description: Tutoring Center Helps children between 6th and 12th grade. Regularly perform instruction sessions on how to apply to college.

Position: Founder, Tutor

UIC Flourish of Chicago (an open source conference)

2009

Description: Volunteer

LANGUAGES

English: Bilingual

Ukrainian: Native Language

Russian: Native Language

COMPUTER SKILLS

Programming: MatLab, Mathematica, Java, C++, Processing, Assembly Languages (MASM and Mini Ide), VHDL

Applications\Platforms: LaTeX, LT-Spice, P-Spice, PhotoShop, DreamWeaver, Office Suits, OmniGrafle

OTHER

IEEE member

U.S. Citizen

REFERENCES

References are available upon request